Frank Riccardi

Senior Software Engineer | AI Research & Model Evaluation

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SUMMARY

Results-driven Senior Software Engineer with 6+ years of professional experience designing, developing, and maintaining scalable software applications, backend services, cloud platforms, APIs, and data-driven systems across healthcare, SaaS, and technology environments. Now specializing in AI model evaluation and mathematical reasoning, including benchmark design, LLM training data development, RLHF/SFT annotation, and statistical analysis of model performance.

Experienced across the full software development lifecycle, including requirements analysis, system architecture, backend engineering, frontend development, database design, API development, testing, deployment, debugging, monitoring, and production support. Strong technical background in Python, JavaScript, TypeScript, Node.js, React, SQL, REST/GraphQL APIs, and cloud technologies.

Hands-on experience with Artificial Intelligence and Machine Learning technologies, including LLM benchmarking, evaluation pipelines, reward and preference data, Python-based data processing, TensorFlow, NumPy, Pandas, SciPy, SymPy, AI API integration, Generative AI, and LLM-powered application functionality.

Experienced in cloud-native and secure software development using AWS, GCP, Azure, Docker, CI/CD, microservices, authentication, access control, and secure API design, with additional experience developing software for healthcare environments requiring careful handling of sensitive data.

EXPERIENCE

Artificial Intelligence Researcher

AngelWatch | Feb 2026 – Present | Contract

- Create mathematics problems in algebra, calculus, linear algebra, and probability to evaluate and train large language models.
- Contribute to benchmark datasets that test multi-step reasoning, following team rubrics and quality guidelines.
- Review model responses for errors such as incorrect calculations and invalid reasoning steps, and document findings for the research team.
- Write step-by-step reference solutions and annotations used for supervised fine-tuning (SFT) and RLHF data.
- Use Python scripts (NumPy, SymPy, Pandas) to check answers and flag incorrect or unclear problems.
- Help create challenging test problems that reveal gaps in model reasoning.

Machine Learning Research Engineer – LLM Benchmarking & Training Data

MoreTranz | Oct 2025 – Dec 2025 | Freelance

- Built automated evaluation pipelines in Python to score LLM outputs on math and reasoning benchmarks, tracking accuracy, consistency, and regressions across model versions.
- Curated and validated training datasets, including deduplication, contamination checks against public benchmarks, and difficulty calibration.
- Built reward-model and preference-ranking datasets by comparing model responses for correctness, rigor, and clarity.
- Applied statistical methods, including confidence intervals, inter-annotator agreement, and significance testing, to make benchmark results reliable.
- Worked with ML engineers to turn evaluation findings into fine-tuning data and prompting strategies that improved reasoning performance.
- Documented benchmark specifications, grading guidelines, and quality standards for annotation teams.

Senior Software Engineer

ATSAP LLC | Sep 2024 – Present | Full-time

- Design, develop, and maintain production-grade software applications and backend systems for clients across multiple industries.
- Architect scalable backend services using Node.js, Express, Python, Firebase, REST APIs, and cloud infrastructure to support reliable application and service integration.
- Build reusable software modules and service-oriented components focused on maintainability, scalability, and extensibility.
- Develop data-processing and automation functionality using Python and JavaScript, including API orchestration, background processing, data transformation, and business logic.
- Integrate third-party APIs, authentication services, payment platforms, messaging services, location services, and cloud-based technologies into enterprise applications.

- Implement secure API endpoints, authentication workflows, authorization controls, data validation, and structured error handling.
- Incorporate AI-powered functionality and intelligent APIs into applications using Python-based components for data processing, automation, and intelligent application features.
- Develop responsive web application interfaces using React, JavaScript, TypeScript, and related frontend technologies.
- Design and optimize database structures and queries to improve application performance, reliability, and scalability.
- Build and maintain cloud deployments using AWS, Firebase, Docker, and CI/CD workflows.
- Diagnose and resolve production issues through debugging, logging, API analysis, database investigation, and application monitoring.
- Collaborate with project managers, designers, developers, and stakeholders to translate business requirements into maintainable technical solutions.
- Participate in software architecture, code reviews, testing, deployment planning, documentation, and continuous application improvement.

Senior Software Engineer

SwivX | Oct 2019 – Dec 2024 | Full-time

- Developed and maintained full-stack software solutions and cloud-connected applications throughout the complete software development lifecycle.
- Designed scalable backend services using Node.js and Python, supporting application logic, data processing, API communication, and cloud integration.
- Built and maintained RESTful APIs for secure communication among frontend applications, backend services, databases, cloud platforms, and third-party systems.
- Developed Python-based data-processing components for transforming, analyzing, and preparing application data for reporting, analytics, and visualization.
- Built responsive web interfaces and dashboards using React, JavaScript, TypeScript, HTML, and CSS, emphasizing usability, maintainability, and performance.
- Designed application workflows connecting user-facing applications with backend services, databases, APIs, and cloud infrastructure.
- Developed and supported healthcare-related software functionality involving cloud-connected applications and connected-device data.
- Supported integrations with Bluetooth Low Energy (BLE) devices and software services for processing and exchanging device-generated data.
- Designed and managed relational and NoSQL databases, including data modeling, query optimization, secure storage, and application-level data access.
- Implemented authentication, authorization, secure API communication, data validation, and access-control mechanisms for applications handling sensitive information.
- Applied secure software engineering practices for healthcare-related applications, including HIPAA-conscious data handling and system architecture.
- Worked with AWS and other cloud technologies to deploy, maintain, monitor, and troubleshoot scalable software services.
- Participated in requirements analysis, architecture planning, code reviews, debugging, testing, deployment, and production support.
- Produced technical documentation, API specifications, integration documentation, and implementation guidance.
- Collaborated with technical and business stakeholders to translate product requirements into scalable and maintainable software solutions.

TECHNICAL SKILLS

- **AI Evaluation & Research:** LLM Benchmarking, Model Evaluation, RLHF, SFT Data Curation, Reward Modeling, Chain-of-Thought Annotation, Adversarial Testing, Statistical Analysis, SymPy
- **Mathematics:** Linear Algebra, Calculus, Probability & Statistics, Number Theory, Discrete Mathematics, Mathematical Proofs
- **Programming Languages:** Python, JavaScript, TypeScript, Java, Kotlin, Swift, Objective-C, PHP, SQL, HTML5, CSS3
- **Backend & Software Engineering:** Node.js, Express.js, NestJS, Django, FastAPI, Laravel, REST APIs, GraphQL, Microservices, Serverless Architecture, Event-Driven Systems, API Integration, Webhooks, Object-Oriented Programming, Design Patterns
- **AI, Machine Learning & Data:** TensorFlow, NumPy, Pandas, SciPy, Machine Learning Fundamentals, Data Preprocessing, AI API Integration, Generative AI, LLM Integration, Prompt Engineering, Intelligent Automation, Data Analysis
- **Frontend Development:** React, Next.js, Angular, Vue.js, Redux, Tailwind CSS, Bootstrap, Material UI, Ant Design, jQuery
- **Databases & Search:** PostgreSQL, MySQL, MongoDB, Redis, Firebase, Elasticsearch, AWS OpenSearch, BigQuery, Sequelize, SQL/NoSQL Database Design, Query Optimization

- **Cloud & DevOps:** AWS, EC2, EKS, Lambda, SNS, SQS, CodeBuild, CodePipeline, GCP, Azure Functions, Firebase Functions, Docker, CI/CD, Vercel, Heroku, Netlify
- **Security & Identity:** OAuth, JWT, SSO, Okta, Auth0, Role-Based Access Control (RBAC), Secure API Design, HIPAA-Aware Development, Data Security
- **Testing & Development Practices:** JUnit, NUnit, PyUnit, Jest, Mocha, Unit Testing, Integration Testing, API Testing, Git, GitHub, GitLab, Agile, Scrum, Code Review, Debugging, Technical Documentation
- **Additional Technologies:** React Native, Flutter, Android, iOS, SendGrid, Strapi

EDUCATION

Marist College — Bachelor of Computer Science (BCompSc) | 2015 – 2019

CERTIFICATIONS

- IBM Full Stack Software Developer Professional Certificate — Coursera | July 17, 2024
- IBM Front-End Developer Professional Certificate — Coursera | July 9, 2024